BAY Version 2 Features and User Flows

Purpose Define the current BAY product, the Version 2 marketplace, and the Melbourne data foundation in one implementation-ready view. Version 2 adds verified private parking supply and native booking without weakening public parking discovery or overstating live accuracy.

Current BAY product

Feature	What exists now	User value
Search and location	Current location, destination search, saved home and work, and return-from-Maps refresh	Avoids restarting a parking search
Map and list	500 m default, 1/2/5 km radii, filters, zoom, recenter and nearby results	Makes nearby options easier to compare
Parking evidence	Distance, restriction, source, freshness, known price and directions	Shows what is known before navigation
Melbourne live layer	Exact sensor identifiers and a five-minute freshness rule	Live status only where the source supports it
Activity and guidance	Saved places, history, support and parking-sign education	Keeps repeat tasks and rules together
Current booking	Verified external operator links where supported	Safe hand-off, but no BAY-owned checkout yet

Current driver flow



Current Melbourne foundation and limits

Available foundation	Recorded scale	Accuracy rule
On-street bays and regional map	29,053 City bays; 46,002 Greater Melbourne mapped bays	A mapped bay is not automatically live, legal or priced
Live sensors	6,324 published sensor rows	Show live only with exact ID and a fresh reading
Restrictions and zones	6,270 Pay Stay restrictions; 935 Pay Stay zones; 926 parking zones	Mixed-sign streets still require exact bay-to-rule validation
Off-street facilities	143 curated commercial car parks	Capacity is not live vacancy; price needs a verified tariff

Current product rule When BAY lacks a fresh, exact live source, the result must say availability on arrival. It must not invent vacancy, price or legal certainty.

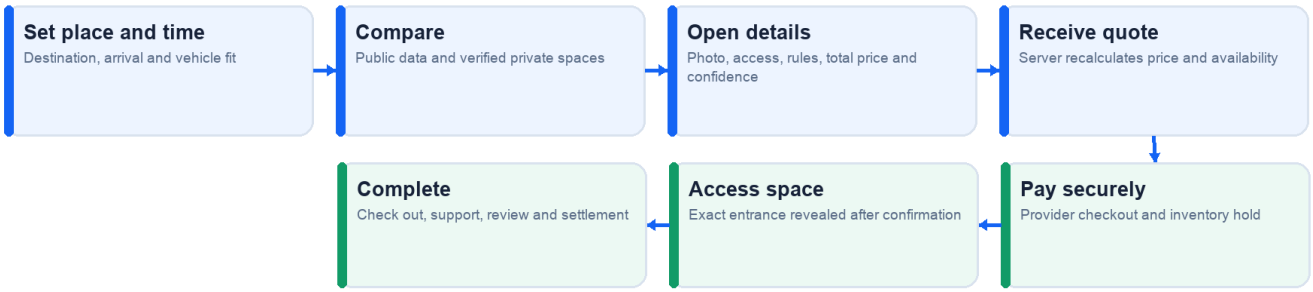
Known gaps carried into Version 2

Gap	Required Version 2 response
Incomplete price coverage	Show a price only when an amount, currency, conditions and effective source are linked unambiguously
Mixed parking signs	Replace broad segment assumptions with exact bay, sign, arrow and time matching where source data permits
Return from navigation	Keep the destination and selected option, then refresh current location and time-sensitive data on resume
No native transaction	Add server quote, inventory hold, secure payment, confirmation, cancellation, refund and payout states

Version 2 Driver Marketplace and Data Foundation

Drivers will search public parking information and verified private spaces together, compare evidence and total price, book a supported space, pay inside BAY and receive exact access instructions only after confirmation.

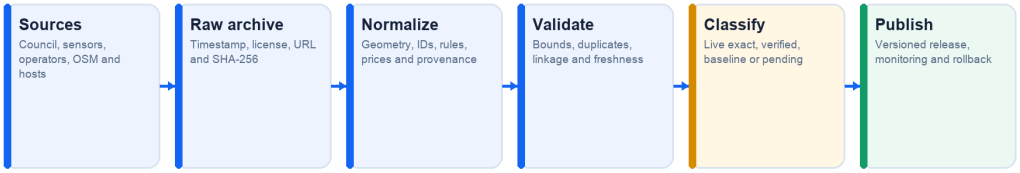
Version 2 driver booking flow



Version 2 feature set

Area	Features included in Version 2
Driver	Location and time search; public/private comparison; photos; vehicle fit; access type; total price; cancellation terms; confidence; directions; support; reviews
Host	Identity and payout verification; map pin; photos; authority and strata evidence; availability; hourly/daily/weekly/monthly prices; calendar; earnings
Marketplace	Listing review; instant or request booking; inventory hold; check-in/out; refunds; disputes; ratings; address privacy
Payments	Server quote; application fee; provider checkout; signed webhooks; idempotency; reconciliation; payout hold and settlement
Data and trust	Source provenance; freshness; confidence tier; moderation; duplicate/public-road checks; field audits; rollback and incident history

Live data and release flow



Melbourne live data cadence and display rule

Data	Update cadence	What BAY may show
Live sensors	Poll about every minute; stale after five minutes	Live status, source and update time only after exact matching
Operator live price or availability	One to five minutes when a licensed API supports it	Provider, effective price and update time
Council restrictions and tariffs	Daily change checks; weekly review; monthly full snapshot	Verified rule or price with source and effective date
OSM and mapped baseline	Monthly rebuild; weekly diffs when stable	Parking location and known attributes; availability on arrival
Host availability and bookings	Transactional and real time	Bookable interval and server-confirmed status

Melbourne coverage standard “Fully covered” means every displayed record has a documented source, license, retrieval time, stable ID, match method, confidence class and release history. It does not mean every physical bay has a sensor or published tariff.

Accuracy display tiers

Tier	Evidence	Display rule
Live exact	Exact source ID and fresh reading	Show live state, source and update time
Verified rule or price	Published council or operator source with an effective date	Show rule or tariff; do not imply live vacancy
Mapped baseline	Source-backed geometry without live or tariff evidence	Show location and known attributes; availability on arrival
Approved host listing	Verified authority, availability and booking state	Show bookable time and total; protect exact access until confirmation

Version 2 Host Operations and Delivery Plan

Private supply can grow only when listing authority, safety, availability, booking state and payments are controlled. Public roads, kerbs, clearways, loading zones, accessible bays and unauthorized common property cannot be listed.

Host listing and payout flow



Booking state control



Key scenarios and impact

Scenario	Version 2 response	Impact to measure
CBD driver before arrival	Compare verified private supply with public evidence, access and total price before directions	Detail-to-quote and completed journey rate
Host lists an apartment bay	Identity, authority, strata evidence, photos, availability, moderation and payout	Verified active supply and time to first booking
Sensor feed becomes stale	Remove the live claim but keep the mapped location with neutral wording	False-vacancy rate and incident recovery time
Two drivers choose one interval	Server inventory lock, quote expiry, idempotency and webhook confirmation	Zero overlapping confirmed bookings
Cancellation or access dispute	Booking timeline, evidence, refund policy and payout hold	Resolution time, refund and chargeback rate

Delivery roadmap and release gates

Phase	Deliverable	Release gate
1 Foundation	Coverage boundary, source registry, confidence model, booking schema and payment test mode	Architecture, policies and rollback approved
2 Supply and trust	Host onboarding, moderation and 50-100 verified spaces in one Melbourne corridor	Authority and field samples pass
3 Closed paid pilot	Quote, payment, access, cancellation, refund and payout	Reconciled transactions; no double booking
4 Melbourne launch	Public/private search, monitoring, monthly coverage report and staffed support	Accuracy, incident and marketplace gates pass

Measures to approve Melbourne launch

Measure	Launch requirement
Data integrity	Every displayed result has provenance and confidence; stale live claims degrade automatically
Booking integrity	No overlapping confirmed bookings; retries cannot create duplicate payment or confirmation
Marketplace trust	Hosts and authority are verified; prohibited-space checks and moderation are operational
Operational readiness	Refund, dispute, incident, privacy and rollback processes are staffed and tested
Measured value	Field accuracy, conversion, completion, cancellation, dispute and repeat-use metrics are reviewed

Version 2 is complete when a verified Melbourne host can publish an authorized space; a driver can find, compare, quote, pay, access, complete or cancel the booking; and BAY can audit every transaction, support both parties and roll back incorrect public data without inventing availability or price.